Alternate Trade Theories

International Economics - T6

Prof. Kalyan Kolukuluri

September 23, 2025

IIM Visakhapatnam

Table of Contents

1. Intra-Industry Trade

2. Gravity Model of Trade

3. Porter's Competitive Advantage

Learning Objectives

Objectives

- Understand alternative theories to explain trade patterns
- Gravity Model
- Intra-Industry Trade

Intra-Industry Trade

Basis for Trade

We will look at Increasing returns to scale and *product differentiation* as basis for trade.

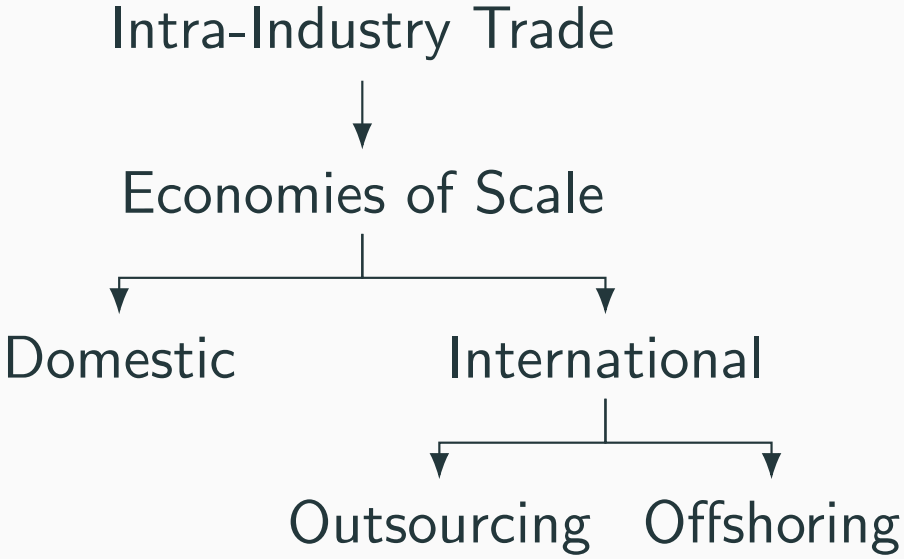
A Paradox in International Trade

Leontief Paradox:

- Until 1950s, trade was believed to be *inter-industry*
- This meant countries would export goods in which it had factor abundance.
- So it was expected, rich countries traded with poor countries.

- But, Wasily Leontief, a Russian-American Harvard economist, observed that the USA exported labor-intensive goods and imported capital goods.
- Industrialized rich countries traded very similar products

Intra-Industry Trade



Product Specialization

Michaely Index:

$$Y_{ij} = \frac{X_{ij}}{\sum_i X_{ij}} - \frac{M_{ij}}{\sum_i M_{ij}}$$

- The index is calculated for one industry 'i'. say, automobile manufacturing in relation to one trading partner 'j', say USA.
- A small (more negative) value of Michaely index indicates no specialization. A high value indicates specialization

Intra Industry Trade - IIT

Grubel - Lloyd Index: A metric of **intra-industry** trade share (IIT share)

$$\text{GL Index} = 1 - \frac{|X_i - M_i|}{X_i + M_i}$$

- X_i Exports of a country in industry i
- M_i Imports of a country in industry i
- GL index $= 1$ if all trade in an intra-industry.
- GL index $= 0$ if all trade in an inter-industry.

Why IIT?

Monopolistic competition: Differentiated products

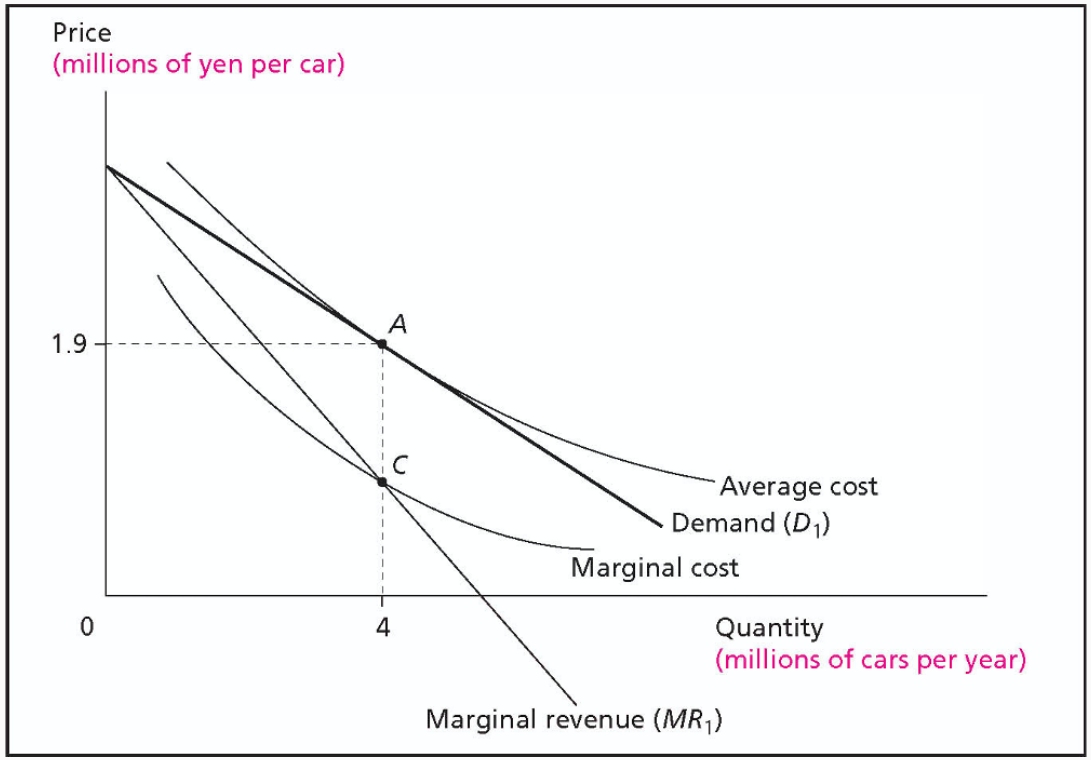


Figure 1: Monopolistic competition

Why IIT?

Monopolistic competition: Differentiated products post trade

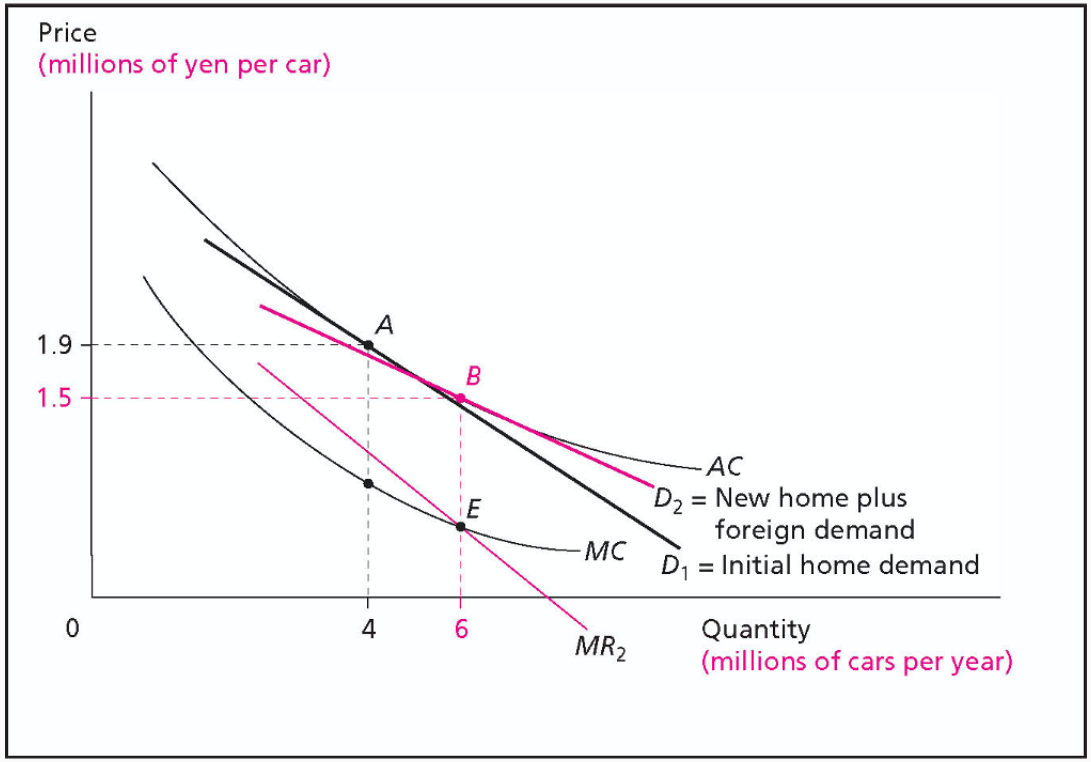


Figure 2: Monopolistic competition

Horizontal & Vertical IIT

Horizontal IIT

When products **differ in their attributes** but do not differ in quality or price. It is assumed that the producers, in the industry, are using same factors of production and same production techniques.

e.g. UK exports LandRover & imports Volkswagen Tiguan

Vertical IIT

When products **differ in terms of quality or price** and it is assumed that the producers, use different factors of production and production techniques.

e.g. Small cars Maruti 800 vs Luxury cars (Bentley)

Horizontal IIT

Figure 3: Ferrari (Export) vs Aston Martin (Import)

Vertical IIT

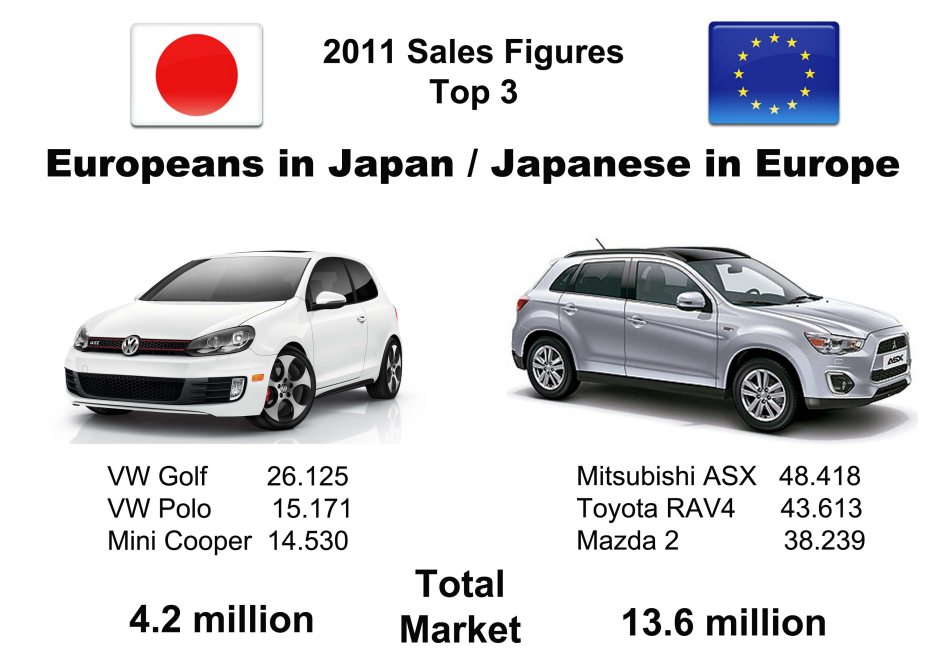


Figure 4: Japanese Compact (Export) vs European Sedan (Import)

Horizontal & Vertical IIT

$$1 - \alpha \leq \frac{UV_{export}}{UV_{import}} \leq \alpha$$

- α is called dispersion factor
- $\alpha \leq 15$ indicates HIIT
- UV_{export} = Value of exports / units of exports
- UV_{import} = Value of imports / units of imports

But why bother? Because, rising VIIT may indicate technological improvement

Gravity Model of Trade

Gravity Model

$$T_{ij} = A\frac{Y_i Y_j}{D_{ij}}$$

1. **Jan Tinbergen** follows Newton.
2. T_{ij} is the total trade flow from origin country i to destination country j
3. Y are the economic size of two country i and j, are usually gross domestic product (GDP) or gross national product (GNP)
4. D_{ij} is the distance between two country i and j. Distance could be cultural or physical, a proxy for transport costs.
5. A is a constant term, that captures any other factor.
6. Indeed US and China are major trading partners.

Porter's Competitive Advantage

Competitive Advantage of Nations

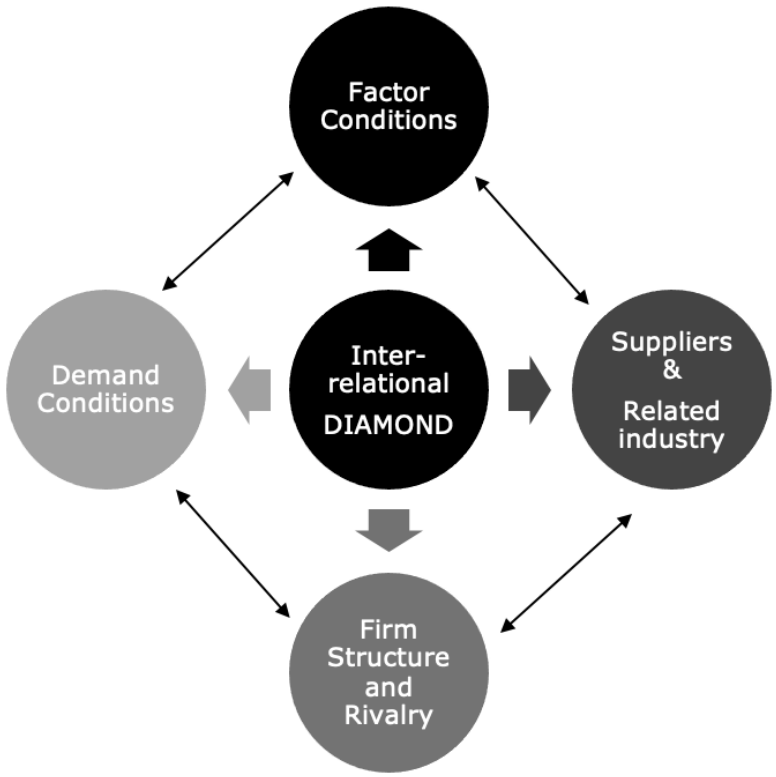
Note: The key word is **competitive** *not* comparative

Figure 5: Michael Porter

Competitive Advantage of Nations

- Factor conditions
- Demand conditions
- Related industry and suppliers
- Firm strategy, structure and rivalry

Porter's Competitive Advantage



A nation's competitiveness depends on the capacity of its industry to *innovate* and *upgrade*.
Companies benefit from having strong domestic rivals, aggressive home-based suppliers, and demanding local customers.

Figure 6: National Competitive Advantage

The End